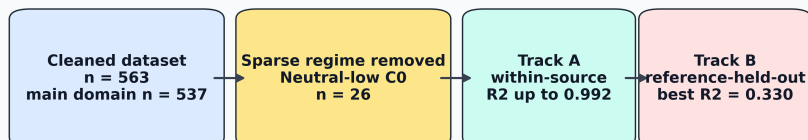


Graphical Abstract

Adsorption ML needs source-aware validation



High interpolation accuracy, applicability domain, and external generalization are different scientific claims.

Highlights

- Source-specific adsorption models reached R^2 up to 0.992.
- Source-origin heterogeneity explained validation sensitivity.
- Row-wise validation overstated source-transfer performance.
- Reference-held-out sensitivity reached $R^2 = 0.330$ in the main domain.
- Trust-score gating reduced capacity-free held-out MAE by 22%.

Source-aware machine learning for AMD-relevant metal adsorption: distinguishing high-accuracy interpolation from cross-source generalization

Yash Bisht^a, Margreth Tadie^b, Susmit Chitransh^b

^a*Department of Mechanical and Industrial Engineering, Indian Institute of Technology Roorkee, Roorkee 247667, Uttarakhand, India*

^b*Department of Chemical Engineering, Stellenbosch University, Stellenbosch, Cape Town, South Africa*

Abstract

Acid mine drainage (AMD) is a persistent water-quality challenge because acidic, sulfate-rich drainage mobilizes metal and metalloid contaminants. Adsorption is widely investigated for AMD-relevant contaminant remediation, and machine learning (ML) can support rapid prediction of removal efficiency; however, reported accuracy can depend strongly on feature choice, applicability-domain coverage, and validation design. This study develops a source-aware and leakage-aware validation framework for adsorption-based metal-contaminant removal-efficiency prediction using a literature-compiled experimental dataset. The contribution is an applicability-domain audit and validation framework, not a claim of a universally deployable adsorption predictor. Source awareness was introduced by tracking the literature reference from which each adsorption record originated. Leakage awareness was addressed by testing whether model performance remained reliable when familiar rows, adsorbents, or entire literature sources were removed from training, and by separating capacity-inclusive process-monitoring models from capacity-free pre-experiment screening mod-

els. The raw dataset contained 566 adsorption records; after deterministic numeric parsing and quality filtering, 563 valid records remained. Because the neutral/low-concentration regime contained only 26 records, primary modeling was restricted to a 537-record domain covering acidic/high-concentration, acidic/low-concentration, and neutral/high-concentration conditions. Track A quantified source-specific interpolation through within-source cross-validation, where XGBoost reached R^2 values of 0.992, 0.974, and 0.960 for three major source groups. Track B tested broader generalization using random row-wise, adsorbent-held-out, and reference-held-out splits. In the restricted primary domain, the full capacity-inclusive LightGBM model achieved $R^2 = 0.806$ under random row-wise validation. Under stricter validation, the best adsorbent-held-out performance was obtained by a capacity-free numeric LightGBM model ($R^2 = 0.532$), while reference-held-out performance reached $R^2 = 0.330$ for a capacity-inclusive LightGBM model and $R^2 = 0.305$ for the best capacity-free model. A source-domain audit showed that source-origin heterogeneity strongly affected validation sensitivity. A capacity-free applicability-domain trust score further reduced reference-held-out MAE from 18.67 to 14.53 when only the highest-trust 40% of predictions were retained, and identified sparse pH- C_0 regions requiring targeted future experiments. These results show that adsorption-ML accuracy can be heavily validation-dependent: high accuracy may indicate interpolation within familiar sources, whereas lower reference-held-out accuracy diagnoses the present gap in external-source transfer.

Keywords: acid mine drainage, adsorption, metal removal efficiency, machine learning, applicability domain, source-aware validation

1. Introduction

Mining activities are a major contributor to water-quality degradation, particularly through the generation of acid mine drainage (AMD). AMD is produced mainly through oxidative dissolution of sulfide minerals and is commonly associated with low pH, high sulfate concentration, and elevated dissolved metal or metalloid concentrations [1]. These conditions can mobilize contaminants such as Fe, Mn, Pb, Cu, Zn, Cr, As, Cd, and Ni, creating long-term risks for receiving water bodies, soils, aquatic ecosystems, and downstream water reuse. Because AMD chemistry varies across mine sites, ore bodies, weathering history, and treatment conditions, robust prediction of contaminant removal remains a difficult environmental-chemical-engineering problem.

Adsorption is widely investigated for AMD-relevant contaminant remediation because it can operate under mild conditions, can use low-cost or waste-derived adsorbents, and can be optimized through pH, adsorbent dose, contact time, temperature, surface chemistry, and initial contaminant concentration. Recent reviews emphasize adsorption as an active research direction for AMD treatment, including carbonaceous, mineral, oxide, biomass-derived, and composite adsorbents [2]. The expansion of experimental adsorption datasets has therefore encouraged the use of machine learning (ML), a subset of artificial intelligence, to predict removal efficiency and accelerate adsorbent or operating-condition screening.

ML is attractive in this setting because adsorption performance depends on nonlinear interactions among solution chemistry, contact time, adsorbent dose, contaminant identity, material surface area, and source-specific synthesis or reporting practices. Data-driven models can capture multivariate trends that are difficult to express with a single empirical equation, and they can provide rapid

screening support once trained on reliable data. However, the validity of an ML model depends not only on the algorithm but also on whether the test data are genuinely independent of the training data and whether the input features are available at the intended stage of deployment.

30 Recent adsorption and environmental-remediation ML studies demonstrate that high predictive accuracy is possible (Table 1). Hafsa et al. modeled heavy-metal adsorption efficiencies using multiple ML algorithms and reported R^2 values in the 0.96–0.99 range under ten-fold cross-validation, with experimental data supplemented by synthetic interpolation [3]. Lu et al. predicted
35 heavy-metal removal by chitosan-based flocculants using random forests and reported $R^2 = 0.9354$, identifying solution pH and flocculant molecular weight as important drivers [4]. Wang et al. used interpretable ML for heavy-metal removal by biochar and reported XGBoost test performance near $R^2 = 0.99$, together with SHAP and partial-dependence analyses to relate removal efficiency to biochar and
40 operating variables [5, 8]. Recent ensemble-learning work on biochar adsorption similarly reported strong XGBoost performance for heavy-metal adsorption efficiency prediction [6]. Within the scope of *Journal of Environmental Chemical Engineering*, interpretable XGBoost modeling has also been applied to heavy-metal removal efficiency in electrokinetic remediation, demonstrating the journal
45 relevance of transparent ML workflows for environmental treatment systems [7].

The central challenge is that high validation accuracy does not necessarily imply reliable generalization to unseen laboratories, adsorbents, contaminant systems, or literature sources. In many adsorption ML studies, performance is estimated using random row-wise validation, where individual experimental records
50 are randomly divided between training and testing. This can be useful for measur-

Table 1: Representative adsorption/remediation ML studies and validation gaps motivating the present work.

Study/domain	Reported strength	Gap addressed here
Heavy-metal adsorption ML [3]	High cross-validation R^2 values, including synthetic interpolation support	Cross-validation can indicate interpolation performance but may not test transfer to unseen literature sources or adsorbent campaigns.
Chitosan-based flocculant removal [4]	Random-forest prediction with $R^2 > 0.93$ and interpretable pH/molecular-weight effects	Strong within-domain prediction does not by itself establish source-held-out or material-held-out generalization.
Biochar heavy-metal adsorption [5, 6]	XGBoost/ensemble models with high test performance and interpretable feature effects	Adsorbent families, feedstocks, and protocols may appear in both train and test sets under row-wise validation.
Electrokinetic remediation ML in JECE [7]	Demonstrates journal relevance of explainable ML for contaminant-removal efficiency	Broader adsorption workflows still need explicit feature-use-case separation and source-aware validation.
PFAS adsorption/partitioning ML [11]	Random-split performance substantially exceeded leave-compound-out transfer	Supports the need to distinguish interpolation from transfer to unseen chemical/source domains.

ing interpolation within a compiled dataset, but it may also place closely related records from the same study dataset, adsorbent series, or operating campaign in both the training and test sets. As a result, the test set can appear independent while still containing familiar source-specific patterns. In ML-based science, such data leakage and validation mismatch can produce overoptimistic estimates of deploy-
55 able performance [9]. Environmental ML best-practice guidance similarly emphasizes careful data splitting, leakage management, randomness assessment, and evaluation design [10]. These concerns are particularly important for adsorption datasets because different studies may use different adsorbent synthesis routes, contaminant species, concentration ranges, analytical protocols, and reporting con-
60 ventions.

Feature definition can further blur the interpretation of reported accuracy. Adsorption capacity and removal efficiency are both derived from concentration changes in many batch adsorption experiments:

$$\text{RE}(\%) = 100 \frac{C_0 - C_e}{C_0}, \quad (1)$$

$$q_e = \frac{(C_0 - C_e)V}{m}, \quad (2)$$

65 where C_0 and C_e are the initial and equilibrium concentrations, V is the solution volume, and m is the adsorbent mass. Because both RE and q_e depend on the concentration change $C_0 - C_e$, adsorption capacity can carry strong information about removal efficiency. A model that uses capacity is therefore useful for process monitoring or post-experiment analysis, where capacity has already been
70 measured or estimated. However, the same model should be treated cautiously for pre-experiment screening, because the capacity of a candidate adsorbent under a

new condition is usually not known before the experiment is performed. Therefore, capacity-inclusive and capacity-free models answer different practical questions and should not be interpreted in the same way. Recent PFAS adsorption modeling has similarly shown that random-split performance can be much stronger than
75 leave-one-compound-out performance, illustrating that interpolation and transfer to unseen chemical domains are distinct tasks [11].

This study addresses these issues by developing a source-aware and leakage-aware validation framework for AMD-relevant metal-contaminant adsorption
80 removal-efficiency prediction across a broad literature-derived dataset spanning 189 normalized source references, 219 adsorbents, and 16 adsorbate classes. Here, source awareness means tracking the literature reference from which each adsorption record originated, rather than treating all rows as fully independent observations. Leakage awareness means testing whether model performance re-
85 mains reliable when familiar information is removed from training, and separating feature sets that may be appropriate after an experiment from feature sets that are suitable before an experiment. The framework is intended to support ML-based mathematical modeling for understanding the combined and individual effects of adsorption variables while avoiding unsupported claims of universal deployment.

90 The framework separates two aspects that are often mixed in adsorption ML studies: what information is available to the model, and how independent the test data are from the training data. Four feature sets were constructed to represent different levels of information availability. A full capacity-inclusive feature set included adsorption capacity and capacity-derived predictors and was interpreted
95 as a process-monitoring or post-experiment analysis model. Three capacity-free feature sets excluded adsorption capacity and were interpreted as pre-experiment

screening models, because the capacity of a candidate adsorbent is usually not known before testing. Among the capacity-free models, one represented adsorbates using ion descriptors, one used normalized adsorbate identity as a categorical label, and one used only numeric operating, material, and engineered process variables as a strict screening baseline.

Two validation tracks were then used to separate interpolation from broader generalization. Track A evaluated source-specific interpolation by training and testing within individual literature sources. This asks whether high R^2 values comparable to those reported in the adsorption ML literature can be reproduced inside known source groups. Track B evaluated broader generalization using increasingly strict split designs. In random row-wise validation, individual records were randomly divided between training and testing. In adsorbent-held-out validation, selected adsorbents were kept completely unseen during training and used only for testing. In reference-held-out validation, all records from selected literature sources were kept unseen during training and used only for testing. Thus, the study asks three practical questions: random validation asks whether the model can interpolate inside the compiled dataset; adsorbent-held-out validation asks whether it can transfer to unseen material names; and reference-held-out validation asks whether it can transfer to unseen literature or source domains. The novelty of this study is therefore not simply a new regression model, but a source-aware and leakage-aware reporting framework that clarifies when high R^2 reflects valid interpolation within known experimental domains and when additional validation is needed before claiming deployment to new adsorbents, studies, or laboratories.

120 2. Materials and methods

2.1. Dataset and target variable

The dataset was compiled from literature-reported adsorption experiments for AMD-relevant metal and metalloid contaminants. Each record corresponds to one adsorption condition with a reported removal efficiency. The raw dataset contained 566 rows and 13 columns. After numeric parsing, missing-value filtering, and constraining removal efficiency to the physically meaningful range 0–100%, 563 records remained. The cleaned dataset included 189 normalized source references, 219 adsorbents, and 16 normalized adsorbate classes. The primary modeling domain excluded the sparse neutral/low-concentration regime, leaving 537 records for model training and testing.

The dataset was intended as a heterogeneous multi-source benchmark for evaluating validation design in adsorption ML, rather than as a systematic meta-analysis of all AMD treatment or heavy-metal adsorption studies. Records were retained when removal efficiency and the required operating or material variables were reported or could be deterministically parsed from tabulated values. Records were excluded when essential numeric variables were missing, when removal efficiency could not be constrained to the physical range of 0–100%, or when adsorbent, adsorbate, or source identity could not be normalized consistently.

The target variable was removal efficiency, RE (%). Input variables included adsorbent name, adsorbate identity, surface area, adsorption capacity, initial concentration, contact time, adsorbent dose, agitation speed, initial pH, temperature, and source reference. The source identifier, denoted Ref_group, was used as a source-origin proxy because explicit laboratory-origin labels were not available for all records. The pH-concentration regime label was not used as a model target

145 or as a model input feature; it was created only for stratified splitting, diagnostic reporting, and applicability-domain definition.

To interpret validation sensitivity, a source-domain audit was performed on the restricted primary domain. This audit quantified the number of records per normalized source, the number of singleton sources, the fraction of rows contributed
 150 by the largest source groups, source-wise RE means and standard deviations, adsorbent and adsorbate diversity within each source, and pH-concentration-regime coverage. A one-way source-label variance diagnostic was also computed as

$$\eta_{\text{source}}^2 = \frac{\sum_g n_g (\bar{y}_g - \bar{y})^2}{\sum_i (y_i - \bar{y})^2}, \quad (3)$$

where g indexes source groups, n_g is the number of records in source g , $\bar{y}_g$ is the mean RE for source g , and $\bar{y}$ is the overall mean RE. This statistic was used only
 155 to diagnose source imprint in the dataset; Ref_group was not used as a predictor in the deployed Track B models.

2.2. Numeric cleaning and categorical normalization

Literature-derived adsorption tables often contain nonuniform numeric formatting. Numeric entries were cleaned using deterministic rules: decimal commas
 160 were converted to decimal points, uncertainty and approximate symbols were removed, multiple values in one cell were reduced to the first numeric value, and nonnumeric text was parsed using regular expressions. Rows with missing values in required numeric variables were removed. Adsorbate names were normalized to reduce duplicates such as elemental and valence-state aliases. Adsorbent names
 165 were grouped into broad families, including carbon-based, mineral/oxide, polymer, biomass, and composite/other categories.

The cleaned data were partitioned into four pH-concentration regimes using pH = 7 as the acidic/near-neutral boundary and the dataset median initial concentration, $C_0 = 50$ ppm, as the low/high concentration boundary (Table 2). The median C_0 is the middle value of the initial-concentration distribution and provides a data-driven threshold for separating lower- and higher-concentration experiments. These regimes were not used as prediction targets and were not passed to XGBoost, LightGBM, or ExtraTrees as predictor variables. Instead, the models predicted RE (%) from the original experimental and material variables, including numerical pH and numerical initial concentration. Regime labels were used only for stratified random splitting and diagnostic analysis, ensuring that train-test splits preserved chemically meaningful coverage across major operating regions. The acidic regimes dominated the dataset, whereas the neutral/low- C_0 regime contained only 26 observations. Because this regime was too small for stable split-wise evaluation, it was excluded from the primary modeling domain and treated as outside the present applicability domain. The excluded regime was not discarded from the dataset audit; it was reported explicitly as an out-of-domain limitation to avoid presenting unstable validation results from a poorly represented operating region.

Table 2: pH-concentration regime distribution in the cleaned dataset.

Regime	Count	Mean pH	Mean C_0 (ppm)	Mean RE (%)
Acidic_HighC0	246	4.51	190.27	70.30
Acidic_LowC0	231	5.12	11.04	74.06
Neutral_HighC0	60	7.65	166.51	68.05
Neutral_LowC0	26	7.47	20.24	81.41

Neutral_LowC0 was excluded from primary modeling because it contained only 26 observations; the other three regimes formed the 537-record primary domain.

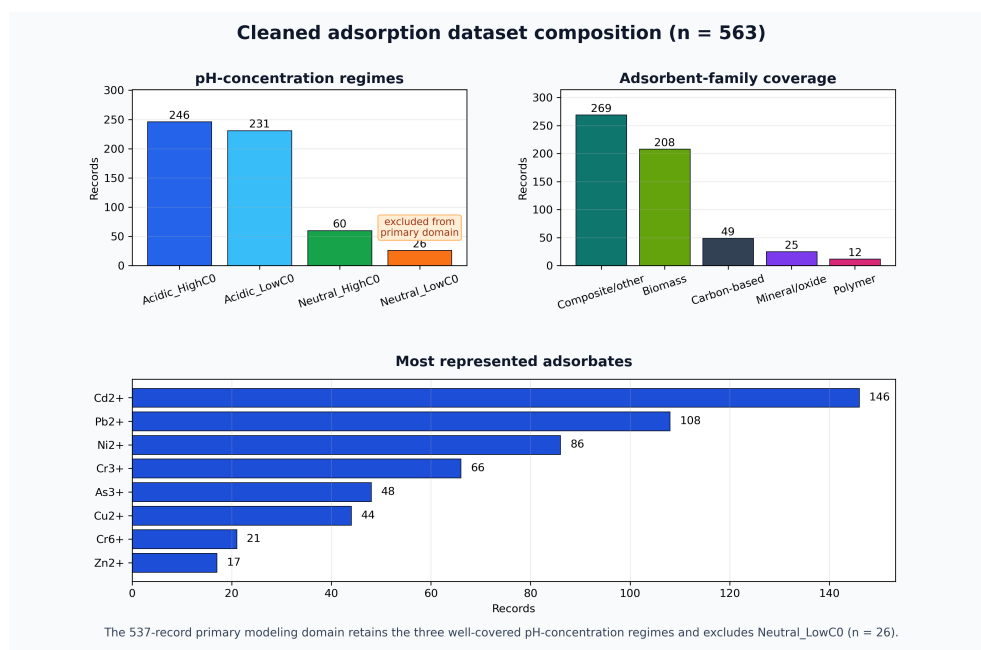


Figure 1: Composition of the cleaned adsorption dataset after deterministic numeric parsing and quality filtering. The sparse neutral/low- C_0 regime was retained in the dataset audit but excluded from the primary 537-record modeling domain.

2.3. Feature sets

185 Four feature sets were constructed to distinguish scientific use cases (Table 3). This separation is important because a model that includes adsorption capacity answers a different question from a model intended for pre-experiment adsorbent screening. Adsorption capacity is closely related to removal efficiency because both are calculated from concentration change during adsorption. In most
190 batch adsorption studies, capacity is not known before an experiment; it is calculated after measuring or estimating equilibrium concentration. Capacity-inclusive models are therefore best interpreted as descriptive, process-monitoring, or post-experiment analysis models. In contrast, capacity-free models use information that is normally available before testing, such as pH, initial concentration, dose, contact time, temperature, surface area, adsorbent family, and adsorbate information.
195 The base experimental inputs were surface area, initial concentration, contact time, dose, agitation speed, initial pH, and temperature. The process-monitoring feature set additionally used adsorption capacity and capacity-derived terms, whereas the screening feature sets deliberately removed these terms.

200 Physics-inspired variables included site density, logarithmic contact time, mixing index, driving-force proxy, and acidity strength:

$$\text{Site Density} = \frac{\text{dose} \times \text{surface area}}{C_0}, \quad (4)$$

$$\text{LogTime} = \log(1 + \text{contact time}), \quad (5)$$

$$\text{Mixing Index} = \text{RPM} \times \text{LogTime}, \quad (6)$$

$$\text{Driving Force} = \frac{C_0}{\text{dose}}, \quad (7)$$

$$\text{Acidity Strength} = |\text{pH} - 7|. \quad (8)$$

Table 3: Feature-set definitions used to separate process-monitoring and screening tasks.

Feature set	Included information	Intended use
Full capacity process model	Operating variables, surface area, adsorption capacity, capacity-derived predictors, engineered process variables, ion descriptors, adsorbent family, and adsorbate identity	Descriptive modeling, process monitoring, or post-experiment analysis
Screening-safe plus ion	Operating variables, surface area, engineered process variables, ion descriptors, and adsorbent family; excludes adsorption capacity and capacity-derived terms	Pre-experiment screening with basic ion chemistry
Screening-safe plus adsorbate one-hot	Operating variables, surface area, engineered process variables, normalized adsorbate identity, and adsorbent family; excludes adsorption capacity and capacity-derived terms	Pre-experiment screening when contaminant identity is known
Screening-safe numeric-only	Operating variables, surface area, and engineered process variables; excludes adsorption capacity, capacity-derived terms, and direct adsorbate identity	Strict screening baseline

These engineered variables were used as heuristic process descriptors rather than mechanistic adsorption-law parameters. They expose the tree models to simple combinations of dose, surface area, concentration, contact time, agitation, and acidity that are commonly relevant to adsorption behavior.

Ion descriptors were added to represent adsorbates through simple aqueous-chemistry properties rather than only through contaminant-name labels. A one-hot adsorbate label identifies the metal species but does not encode chemical similarity between ions. Descriptors such as charge, ionic radius, hydrated radius, ionic potential, and hydration ratio allow the model to partially capture differences in ion size, charge density, and hydration behavior that can influence adsorption.

The full model therefore used the following ML inputs: seven operating/material numeric variables; adsorption capacity, capacity-loading ratio, and thermo-capacity; five engineered process descriptors; six ion descriptors; normalized adsorbate identity; and broad adsorbent family. The capacity-free models retained the operating/material variables and engineered descriptors but removed adsorption capacity, capacity-loading ratio, and thermo-capacity. Among the capacity-free models, one retained ion descriptors, one used normalized adsorbate identity, and the strictest model used numeric variables only.

Capacity-derived predictors were included only in the capacity-inclusive process model and were excluded from all screening-safe models. They were defined as

$$\text{Capacity Loading Ratio} = \frac{q_e}{C_0}, \quad (9)$$

$$\text{Thermo Capacity} = q_e T, \quad (10)$$

where q_e is adsorption capacity, C_0 is initial concentration, and T is temperature.

2.4. Machine-learning models

XGBoost was selected as the primary model because gradient-boosted trees are effective for nonlinear tabular datasets and are widely used in adsorption ML studies [12, 13]. LightGBM was included as an additional gradient-boosting comparator because preliminary model development indicated similar row-wise interpolation performance [14]. Extremely randomized trees (ExtraTrees) were included as a comparator ensemble model because random forest-style tree ensembles provide robust nonlinear baselines for tabular prediction [15, 16]. A mean dummy regressor was included to quantify improvement beyond a no-skill baseline.

Categorical variables were one-hot encoded with unknown-category handling, while numerical variables were passed directly to the tree models. For each split, categorical encoding and model fitting were performed within the training fold only, and the fitted preprocessing pipeline was then applied to the corresponding test fold. Unknown categories appearing only in test splits were ignored by the encoder rather than fitted using test information. This ensured that preprocessing did not introduce information from the test set into the training process.

XGBoost used 140 estimators, learning rate 0.05, maximum depth 4, subsample 0.85, column subsample 0.85, ℓ_1 regularization 0.1, and ℓ_2 regularization 1.0. The standard LightGBM model used 300 estimators, learning rate 0.05, maximum depth 5, 31 leaves, minimum child samples 15, subsample 0.85, column subsample 0.85, ℓ_1 regularization 0.1, and ℓ_2 regularization 0.5. ExtraTrees used 120 estimators, maximum depth 10, and minimum leaf size 2.

Two additional LightGBM sensitivity configurations were evaluated because preliminary model development had produced strong single-split LightGBM performance. These configurations were not used to claim a newly optimized al-

gorithm; they were included as sensitivity checks to test whether the validation conclusions were stable across reasonable gradient-boosting complexity settings. The conservative configuration used 600 estimators, learning rate 0.03, maximum depth 5, 20 leaves, minimum child samples 25, subsample 0.80, column subsample 0.80, ℓ_1 regularization 0.5, and ℓ_2 regularization 1.0. The expanded configuration
250 used 500 estimators, learning rate 0.035, maximum depth 6, 40 leaves, minimum child samples 10, subsample 0.90, column subsample 0.90, ℓ_1 regularization 0.05, and ℓ_2 regularization 0.2. Predictions were clipped to the physically meaningful RE range of 0–100% before metric calculation. This clipping was applied only to
255 final model predictions and did not affect training targets.

2.5. Validation design

Two validation tracks were used (Fig. 2). Track A evaluated source-specific interpolation. Although the cleaned dataset contained 189 normalized source references, most references contributed only a small number of rows. Therefore,
260 source-specific 5-fold cross-validation was reported only for references with at least 20 usable records and sufficient variation in removal efficiency, using shuffled folds with a fixed random seed. Here, “5-fold” refers to the number of cross-validation folds within each eligible source group, not to the number of source groups reported. This tests whether high R^2 values are achievable within
265 a known experimental source, which is similar to the validation setting of many high-performance adsorption ML studies.

Track B evaluated global generalization using three validation protocols within the restricted primary domain. The random row-wise split used an 80/20 train-test split stratified by the three retained pH-concentration regimes, so each split re-
270 tained acidic/high- C_0 , acidic/low- C_0 , and neutral/high- C_0 observations. This split

answers the in-dataset interpolation question: can the model predict held-out rows when related rows from the same sources may remain in training? This stratification affected only how rows were assigned to training and test sets; the regime label itself was not included in the fitted model. The adsorbent-held-out split used group splitting so that exact adsorbent material names in the test set were absent from the training set; all rows for a held-out adsorbent identity were assigned to the test split. This split answers the cross-material question: can the model transfer to unseen material names when measured descriptors such as surface area, operating conditions, pH, dose, and concentration are available? It does not claim prediction for a completely undescribed material from name alone. The reference-held-out split used group splitting so that entire references/sources were absent from the training set, making it the stricter proxy for transfer across literature sources, experimental protocols, adsorbent synthesis routes, concentration windows, contaminant distributions, and reporting practices. This split answers the external-source question: can the model transfer to unseen literature/source domains? Because all data were literature-derived, reference-held-out validation was used as a proxy for external-source validation and should not be interpreted as a substitute for future prospective laboratory validation. Each Track B protocol was repeated over five random seeds. Metrics included R^2 , root mean squared error (RMSE), mean absolute error (MAE), and prediction bias.

2.6. *Applicability-domain trust score and experiment-priority map*

To convert the capacity-free reference-held-out result into a practical screening workflow, an applicability-domain trust score was computed for the best capacity-free reference-held-out model, namely the screening-safe numeric-only ExtraTrees model. The trust score was not used to train or refit the regression model, and it

did not use RE, adsorption capacity, or prediction error. It was used only after prediction to rank whether a capacity-free prediction was well supported by the training domain or should instead be treated as a candidate for new experimental measurement.

300 The score combined four pre-experiment support terms. First, nearest-neighbor distance support measured how close each test record was to the training distribution in a standardized capacity-free descriptor space containing operating variables, material descriptors, engineered process variables, and ion descriptors. Second, physical-envelope support measured the fraction of core variables lying
305 within the 5th–95th percentile range of the training set. Third, categorical support checked whether adsorbent family, adsorbate group, and pH-concentration regime were represented in the training set. Fourth, source-neighborhood diversity measured how many distinct source references appeared among the ten nearest training neighbors. The final trust score used weights of 0.45, 0.20, 0.20, and
310 0.15 for these four terms, respectively. These weights intentionally prioritized geometric proximity while still penalizing out-of-envelope conditions, unseen categories, and narrow source support.

A separate future-experiment priority map was constructed by binning the full cleaned dataset across pH and initial concentration. The pH bins were $\text{pH} < 4$, $4 \leq$
315 $\text{pH} < 5.5$, $5.5 \leq \text{pH} < 7$, and $\text{pH} \geq 7$; the C_0 bins were $C_0 \leq 10$, $10 < C_0 \leq 50$, $50 < C_0 \leq 100$, and $C_0 > 100$ ppm. Cells with fewer than 30 records were flagged as high-priority regions for future experiments, because they represent operating spaces where capacity-free models are likely to extrapolate rather than interpolate.

Source-aware validation framework for adsorption removal-efficiency prediction

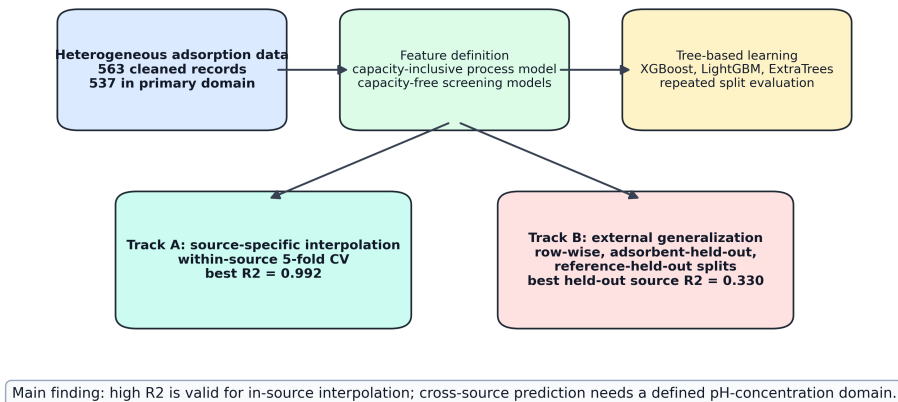


Figure 2: Workflow of the proposed source-aware adsorption-ML validation framework. Track A measures source-specific interpolation, whereas Track B measures progressively stricter global generalization.

3. Results

320 The results are reported in the same order as the validation framework:
first, source-specific interpolation within coherent literature sources; second,
repeated random row-wise prediction in the restricted primary domain; third,
stricter adsorbent-held-out and reference-held-out transfer; fourth, comparison
of capacity-inclusive and capacity-free feature sets; and fifth, reliability-aware
325 capacity-free screening. This ordering separates interpolation accuracy from
broader external-source generalization.

3.1. Source-specific models reproduced high adsorption-prediction accuracy

Track A showed that high removal-efficiency prediction accuracy is achiev-
able when models are trained and evaluated within individual source groups. The
330 source groups in Table 4 are not the only sources in the dataset; they are the clean
bibliographic source groups that met the minimum row-count and target-variability
criteria and produced stable positive within-source cross-validation performance.
The full capacity process model achieved $R^2 = 0.9916$ for the source group Shen
et al. (2017b), $R^2 = 0.9735$ for Gao et al. (2019), and $R^2 = 0.9601$ for Zama
335 et al. (2017). These results are in the range commonly reported in adsorption
ML studies under row-wise or within-domain validation [3, 5, 4]. They are also
chemically plausible because the dominant source groups are internally coherent
adsorption campaigns rather than arbitrary row collections: Shen et al. studied
Ni(II) adsorption on wheat-straw-pellet and rice-husk biochars, Gao et al. inves-
340 tigated Cd(II) adsorption mechanisms on rice-straw and sewage-sludge biochars,
Zama et al. studied Pb(II), Cd(II), and As(III) sorption/desorption across biochars,
and Cui et al. focused on Cd(II) removal by *Canna indica*-derived biochar [17, 18,

19, 20]. Two additional ScienceDirect-indexed URL-bucket groups met the row-count threshold but were not treated as clean source-specific Track A groups because their reference metadata indicated mixed or unresolved bibliographic grouping. They are retained in the benchmark audit trail, but not used to support the high-interpolation claim. This reinforces the central point that source-specific predictability should be evaluated only for coherent source groups and should not be summarized by a single global accuracy value.

Table 4: Track A source-specific 5-fold cross-validation results for the four coherent source groups with stable positive within-source predictability. Complete Track A benchmark outputs, including unresolved ScienceDirect URL-bucket groups, are retained in the submitted CSV files for auditability.

Source group	Rows	R^2	RMSE	MAE
Shen et al. (2017b)	65	0.9916	3.20	2.33
Gao et al. (2019)	48	0.9735	2.97	2.14
Zama et al. (2017)	91	0.9601	7.84	3.56
Cui et al. (2016a)	24	0.8936	6.65	4.56

Capacity-free source-specific models also performed well in selected source groups. For example, the source-specific model for Zama et al. (2017) achieved $R^2 = 0.9250$ using the screening-safe ion feature set, while Shen et al. (2017b) achieved R^2 values above 0.91 using capacity-free screening feature sets. This indicates that high source-specific performance is not solely a capacity-inclusion artifact, although capacity-inclusive models generally gave the strongest interpolation performance.

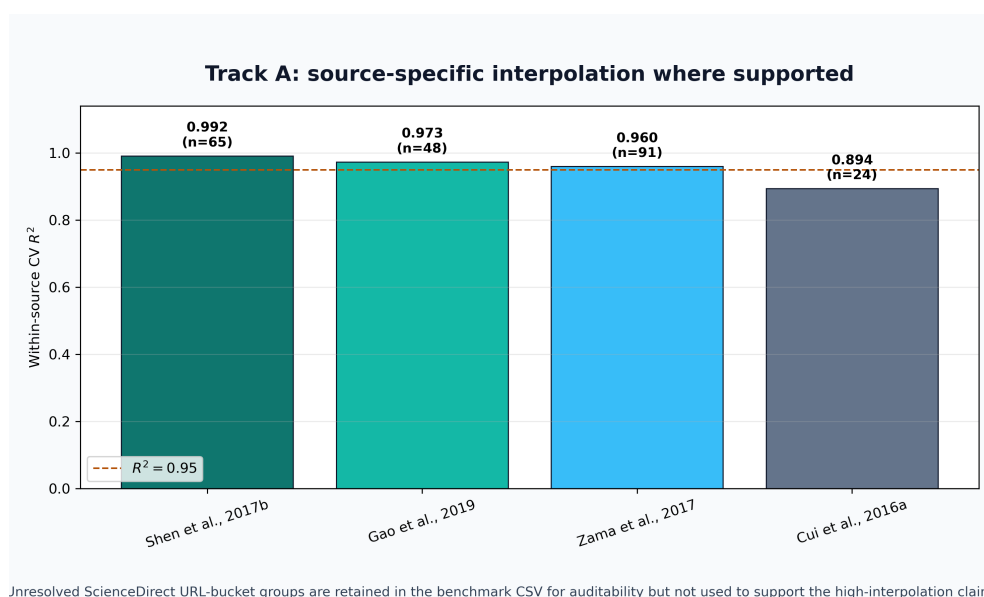


Figure 3: Track A source-specific interpolation performance. Several individual source-group models achieved R^2 values near or above 0.95, while other source groups remained more difficult.

3.2. Random row-wise validation produced moderate-to-good global performance

Under repeated random row-wise splits in the restricted primary domain, the
 360 full capacity process model achieved $R^2 = 0.8064$ and RMSE = 12.94 with Light-
 GBM, while the corresponding XGBoost model achieved $R^2 = 0.7985$ and RMSE
 = 13.22. The screening-safe plus adsorbate one-hot XGBoost model achieved
 $R^2 = 0.7283$ and RMSE = 15.29, while the screening-safe plus ion XGBoost
 model achieved $R^2 = 0.7265$ and RMSE = 15.34. Thus, even without adsorption
 365 capacity, the model retained useful predictive ability under in-dataset interpolation
 (Table 5).

Table 5: Restricted-domain random row/regime split performance after excluding the sparse neutral/low- C_0 regime. Values are means over five repeated splits.

Feature/model	R^2	RMSE	MAE
Full capacity LightGBM	0.8064	12.94	8.08
Full capacity XGBoost	0.7985	13.22	8.44
Full capacity ExtraTrees	0.7730	14.02	9.02
Screening-safe plus adsorbate XGBoost	0.7283	15.29	10.84
Screening-safe plus ion XGBoost	0.7265	15.34	10.88

These results support the use of ML for in-domain adsorption prediction.
 They also explain the strong preliminary LightGBM results obtained during
 model development. Earlier LightGBM reports used a single random/regime-
 stratified split and included capacity-related predictors, giving test $R^2 = 0.847$ and
 370 RMSE = 12.05; in the present repeated-split benchmark, individual LightGBM
 random splits similarly reached $R^2 = 0.8499$ for the standard configuration and

$R^2 = 0.8570$ for the conservative sensitivity configuration. However, the five-split means were lower ($R^2 = 0.8064$ and $R^2 = 0.7983$, respectively), showing that
375 the earlier value was a valid favorable row-wise interpolation result rather than a stable estimate of source-held-out generalization. Random row-wise validation alone is therefore not sufficient evidence for deployment on unseen sources.

The two capacity-free XGBoost screening models differed in how the adsorbate was represented. The screening-safe plus adsorbate model used normalized
380 adsorbate identity as a categorical one-hot feature, allowing the model to learn contaminant-specific effects directly. The screening-safe plus ion model used simple aqueous-ion descriptors, including charge, ionic radius, hydrated radius, ionic potential, and hydration ratio, instead of the direct contaminant label. The similar row-wise performances of these two models, $R^2 = 0.7283$ and $R^2 = 0.7265$,
385 respectively, indicate that basic ion-chemistry descriptors captured much of the information provided by explicit adsorbate labels in the random-split setting.

3.3. Cross-material and cross-source validation tested broader transfer

The validation protocol strongly affected performance, which is expected for a heterogeneous dataset spanning many source references, adsorbent identities, and
390 metal ions. In the adsorbent-held-out protocol, all experiments associated with selected adsorbent material names were removed from the training set and used for testing. This is different from holding out adsorbates: the held-out groups are adsorbent materials, not contaminant ions. Under this cross-material test, the best capacity-free result was obtained by the screening-safe numeric expanded-
395 LightGBM model, with $R^2 = 0.5323$ and RMSE = 19.35. The screening-safe numeric XGBoost model achieved $R^2 = 0.5199$, while the conservative full-capacity LightGBM model achieved $R^2 = 0.5258$. These results indicate moderate but

imperfect transfer to unseen adsorbent identities when measured descriptors and operating conditions are available.

400 Reference-held-out validation was the most important broad-source stress test because every row from selected literature references was kept outside training. The model was therefore evaluated on source domains whose experimental protocol, adsorbent preparation, concentration window, contaminant distribution, and reporting pattern were not represented by the same reference during training. 405 This validation is closer to the practical question of whether a literature-trained model transfers to a new study. It was more difficult than row-wise validation but improved markedly after removing the sparse neutral/low- C_0 regime. The best reference-held-out sensitivity result was obtained by the full-capacity expanded-LightGBM model, with $R^2 = 0.3296$ and RMSE = 22.88. Because 410 this configuration includes capacity-related predictors, it should be interpreted as a process-monitoring or post-experiment result rather than as a pre-experiment screening result. The best capacity-free reference-held-out result remained the screening-safe numeric-only ExtraTrees model, with $R^2 = 0.3054$ and RMSE = 23.54. These results remain below row-wise performance, but they show 415 that defining a better-covered applicability domain produces more meaningful external-source validation for a broad, multi-source adsorption dataset.

These results show that removal-efficiency prediction is strongly source-dependent. The same data and model family can produce high R^2 under source-specific interpolation, moderate-to-good R^2 under random global validation, and lower but still informative R^2 under source-held-out validation when 420 the applicability domain is restricted to sufficiently represented pH-concentration regimes.

Table 6: Restricted-domain Track B generalization performance after excluding the sparse neutral/low- C_0 regime. Values are means over five repeated splits.

Validation	Feature/model	R^2	RMSE	MAE
Holdout adsorbent	Screening-safe numeric expanded LightGBM	0.5323	19.35	14.55
Holdout adsorbent	Full capacity conservative LightGBM	0.5258	19.19	13.00
Holdout adsorbent	Screening-safe numeric-only XGBoost	0.5199	19.61	15.15
Holdout reference/source	Full capacity expanded LightGBM	0.3296	22.88	17.67
Holdout reference/source	Screening-safe numeric-only ExtraTrees	0.3054	23.54	18.67
Holdout reference/source	Full capacity standard LightGBM	0.2820	23.78	18.67

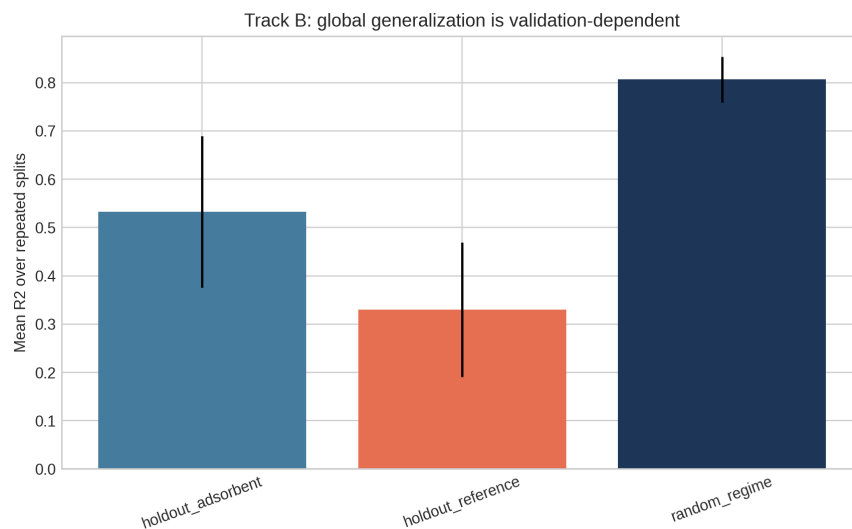


Figure 4: Track B validation comparison. Row-wise validation, adsorbent-held-out validation, and reference-held-out validation represent increasingly strict generalization tasks.

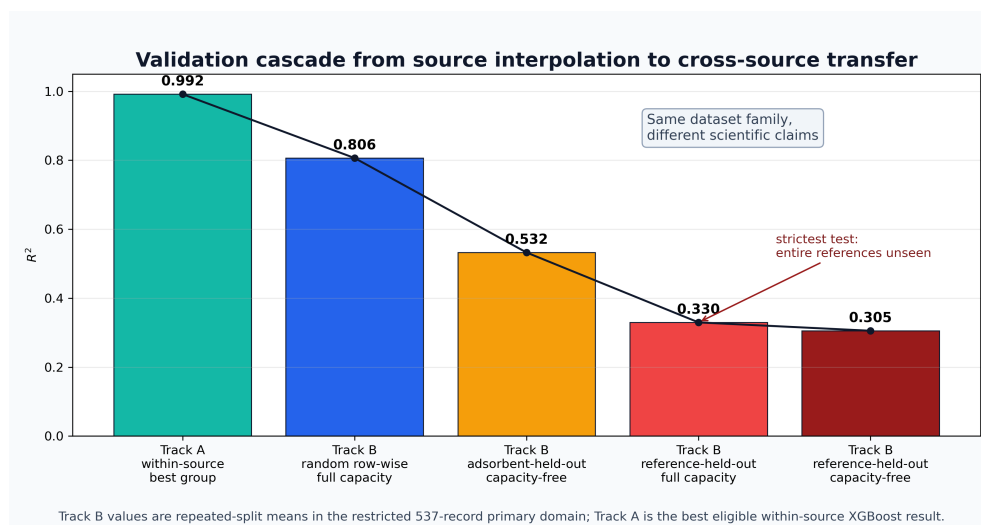


Figure 5: Validation cascade from source-specific interpolation to stricter global transfer. The figure summarizes why high R^2 values near 0.95 are achievable within known sources while reference-held-out prediction is a more difficult external-source task.

3.4. *Applicability-domain gating identified lower-risk capacity-free predictions*

Because capacity-free models are the relevant models for pre-experiment
425 screening, the best capacity-free reference-held-out configuration was further
evaluated with the applicability-domain trust score. Without abstention, the
screening-safe numeric-only ExtraTrees model achieved the same five-split mean
reported above: $R^2 = 0.3054$, RMSE = 23.54, and MAE = 18.67. When pre-
430 dictions were ranked by pre-experiment trust score, error decreased as low-trust
predictions were removed from the reported screening set (Fig. 6). Retaining only
the highest-trust 40% of reference-held-out predictions reduced MAE from 18.67
to 14.53 and RMSE from 23.54 to 19.73, corresponding to reductions of 22.2%
and 16.2%, respectively.

The same pattern was visible by trust quartile. The lowest-trust quartile had
435 MAE = 29.44, whereas the highest-trust quartile had MAE = 14.26, giving a 51.6%
lower absolute error for high-trust predictions. The retained high-trust subset did
not maximize R^2 ; at 40% coverage, R^2 was 0.126 because the retained subset had
a narrower response distribution and R^2 is normalized by target variance. There-
fore, the abstention analysis should be interpreted through absolute-error reduction
440 and reliability ranking, not as an attempt to inflate R^2 . Practically, this means a
capacity-free adsorption model should not be forced to predict every new condi-
tion. It can instead report a prediction only when the new condition is sufficiently
supported by the training domain and flag low-trust cases as priorities for new ex-
periments.

445 3.5. *Source-domain heterogeneity explained validation sensitivity*

The decrease from source-specific or row-wise performance to reference-held-
out performance was not simply an algorithmic failure; it reflected the structure

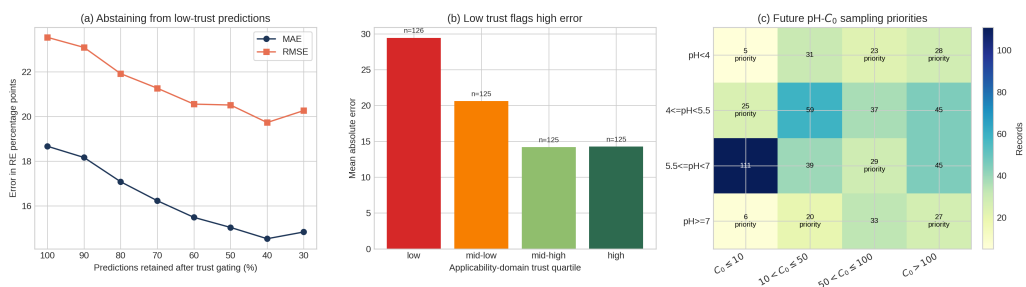


Figure 6: Reliability-aware capacity-free screening and future experiment design. (a) Applicability-domain trust gating reduced MAE and RMSE for the reference-held-out capacity-free ExtraTrees model as low-trust predictions were abstained from. (b) Low-trust predictions had substantially higher absolute error than high-trust predictions. (c) Sparse pH–C₀ regions in the full cleaned dataset identify priority operating windows for future adsorption experiments.

of the literature-compiled dataset. In the 537-record primary domain, the records were distributed across 180 normalized source references, but 142 of those sources contributed only one record and the median source size was one row (Table 7). Only six sources contributed at least 20 records, yet those six sources accounted for 52.7% of the primary-domain rows. Therefore, a random row-wise split can place closely related experiments from the same source campaign in both training and testing, whereas reference-held-out validation deliberately removes that source-specific support.

This audit identifies the hidden assumption behind the performance drop. Row-wise validation implicitly assumes that individual rows are exchangeable samples from one common adsorption distribution. The compiled dataset violates that assumption: source groups differ in contaminant identity, adsorbent synthesis route, pH and concentration window, measured material descriptors, analytical protocol, and reporting convention. The source-label variance diagnostic attributed 42.4%

Table 7: Source-domain heterogeneity diagnostics for the 537-record primary modeling domain.

Diagnostic	Value
Primary-domain records	537
Normalized source references	180
Singleton source references	142
Median records per source reference	1
Sources with at least 20 records	6
Rows contributed by six largest source groups	52.7%
Overall RE mean $\pm$ standard deviation	71.66 $\pm$ 29.92 percentage points
Standard deviation of source-wise RE means	19.26 percentage points
Median within-source RE standard deviation	7.37 percentage points
Source-label variance diagnostic, η^2_{source}	0.424

The source-label variance diagnostic is descriptive and was not used as a model input. Because many references were singletons, η^2_{source} should be interpreted as evidence of source imprint rather than as a causal partition of adsorption mechanisms.

of the RE variance to source grouping, and the spread of source-wise mean RE values was larger than the median within-source spread. Thus, reference-held-out validation is not merely a harsher score; it tests whether a model can transfer when
465 source-specific experimental context is no longer shared between training and testing.

3.6. *Capacity-inclusive and capacity-free models answered different questions*

The full capacity process model performed best for random row-wise validation and several source-specific models. This is expected because adsorption capacity
470 carries strong information about adsorption performance. However, adsorption capacity is derived after measuring concentration change and is usually not available before testing a candidate adsorbent under a new condition. Therefore, capacity-inclusive results should be interpreted as descriptive, process-monitoring, or post-experiment analysis results. Capacity-free models are more appropriate for pre-
475 experiment screening and adsorbent-design guidance because they exclude adsorption capacity and capacity-derived predictors. In the restricted-domain adsorbent-held-out and reference-held-out tests, capacity-free numeric models were competitive or superior in several cases. This should not be interpreted as adsorption capacity being uninformative; rather, it suggests that capacity-related predictors can
480 improve interpolation while also encoding source- or condition-specific relationships that do not always transfer cleanly across held-out materials or references.

4. Discussion

4.1. *Why near-0.95 R^2 is achievable but validation-dependent*

The Track A results explain why many adsorption ML studies report R^2 values
485 near or above 0.95. Within a single source, adsorbent family, contaminant sys-

tem, or experimental protocol, the response surface can be learned accurately by tree-based models. In this setting, high R^2 is scientifically plausible and not automatically invalid. The present results reproduce that behavior, with source-specific R^2 values of 0.960–0.992 for three major source groups.

490 However, Track B shows that this performance should not be interpreted as universal adsorption prediction. When sources are held out, the model must transfer across experimental protocols, adsorbent synthesis methods, contaminant distributions, concentration ranges, and reporting conventions. Under that validation setting, performance fell below row-wise performance even after excluding
495 the sparse neutral/low- C_0 regime. The result is consistent with recent adsorption-related work showing that random-split performance can be substantially higher than leave-compound-out performance [11].

The lower reference-held-out R^2 should therefore be read as a diagnostic finding rather than only a modeling weakness. It is not evidence that ML is unusable
500 for adsorption; instead, it reveals the gap between interpolation within familiar source domains and transfer to unseen literature/source domains. It also indicates that important context is missing from the current descriptor set. Likely missing variables include pH at point of zero charge, pore volume, pore-size distribution, surface functional groups, zeta potential, cation-exchange capacity, feedstock
505 and pyrolysis details for biochars, aqueous speciation, ionic strength, competing ions, and laboratory protocol metadata. Without such descriptors, the model can learn source-specific patterns under interpolation but has limited information for explaining why a new source behaves differently.

4.2. Feature availability changes the scientific claim

510 The comparison between capacity-inclusive and capacity-free feature sets shows that feature availability changes the meaning of the prediction task. Adsorption capacity carries strong information about removal efficiency because both are derived from concentration change during adsorption. Capacity-inclusive models are therefore useful for descriptive modeling, process monitoring, and
515 post-experiment analysis, but they should not be interpreted as purely prospective adsorbent-screening models unless capacity is already known or estimated independently.

In contrast, capacity-free models exclude adsorption capacity and capacity-derived terms, making them more appropriate for pre-experiment screening. These
520 models rely on information that is typically available before testing, such as pH, initial concentration, contact time, dose, temperature, surface area, adsorbent family, and adsorbate information. The similar row-wise performance of the adsorbate-label and ion-descriptor screening models indicates that simple ion-chemistry descriptors can capture much of the information provided by explicit contaminant
525 labels in the random-split setting. The competitive performance of capacity-free numeric models under held-out validation further suggests that simpler descriptor sets may sometimes transfer more robustly across materials or sources, even if capacity-inclusive models remain stronger for within-domain interpolation.

4.3. Experimental design needed to improve capacity-free screening

530 If the intended use is pre-experiment adsorbent screening, capacity-free models are the more defensible modeling pathway because they use only information that can be known before measuring the final concentration change. The lower held-out R^2 values in the present study therefore indicate what future experiments

must measure more systematically. A capacity-free model can improve only if the
535 variables that control adsorption are available as inputs before the adsorption test,
rather than being indirectly supplied through post-experiment capacity.

Future datasets should therefore be designed as prospective ML benchmarks
rather than assembled only from optimized adsorption case studies. Each adsor-
bent should be characterized before adsorption using descriptors such as BET
540 surface area, pore volume, pore-size distribution, pH at point of zero charge, zeta
potential, cation-exchange capacity, elemental composition, surface functional
groups from FTIR/XPS, feedstock, pyrolysis or activation temperature, and mod-
ification chemistry. Solution chemistry should be recorded with similar detail,
including initial pH, initial contaminant concentration, sulfate concentration,
545 ionic strength, competing ions, alkalinity, redox condition where relevant, and
whether the water is synthetic single-ion solution, synthetic AMD, or real AMD.
These descriptors would give a capacity-free model direct chemical information
that is currently hidden inside source-specific patterns. The trust-score result
supports this view: high-error predictions were not randomly distributed but were
550 concentrated in weaker applicability-domain regions.

The experimental matrix should also be balanced across the operating space.
Instead of reporting mainly high-performing or optimized conditions, future stud-
ies should deliberately sample low, medium, and high initial concentrations; acidic
and near-neutral pH; multiple adsorbent doses; several contact times; and multi-
555 ple metal ions or mixed-metal AMD-relevant systems. The pH- C_0 priority map
identified several high-priority cells with fewer than 30 records, including pH < 4
with $C_0 \leq 10$ ppm (5 records), pH ≥ 7 with $C_0 \leq 10$ ppm (6 records), pH ≥ 7
with $10 < C_0 \leq 50$ ppm (20 records), pH < 4 with $50 < C_0 \leq 100$ ppm (23

records), $4 \leq \text{pH} < 5.5$ with $C_0 \leq 10$ ppm (25 records), $\text{pH} \geq 7$ with $C_0 > 100$ ppm (27 records), $\text{pH} < 4$ with $C_0 > 100$ ppm (28 records), and $5.5 \leq \text{pH} < 7$ with $50 < C_0 \leq 100$ ppm (29 records). These regions should be filled using multiple adsorbent families and contaminant ions, rather than one optimized adsorbent-condition pair, so that future models learn transferable response structure instead of source-specific shortcuts. Replicate experiments should be included to estimate experimental noise, and a subset of adsorbents, contaminants, and laboratory batches should be reserved as blinded holdout groups before model fitting. This design would reduce random-split optimism and make adsorbent-held-out, contaminant-held-out, laboratory-held-out, and reference-held-out validation more stable.

In this workflow, adsorption capacity should still be measured and reported, but it should be treated as an outcome or diagnostic variable rather than a predictor for screening. A practical future protocol would first train a capacity-free model using only pre-test descriptors to propose promising adsorbent-condition combinations, report an applicability-domain trust score for each proposed condition, experimentally prioritize low-trust but environmentally important regions, measure both RE and capacity, and finally use the new results to update the model through active learning. Such an iterative design would improve R^2 not by hiding outcome-derived information in the feature set, but by expanding the chemically relevant input space and reducing extrapolation to poorly described sources.

4.4. Relation to existing adsorption ML literature

The proposed framework complements rather than contradicts previous high-accuracy adsorption ML studies. Hafsa et al. demonstrated that tree-based algorithms can achieve high removal-efficiency prediction accuracy for

heavy-metal adsorption datasets under repeated cross-validation [3]. Wang et al. showed that XGBoost, SHAP, and partial-dependence analyses can identify interpretable drivers of heavy-metal removal by biochar [5]. Lu et al. showed that random forests can predict heavy-metal removal by chitosan-based flocculants with R^2 above 0.93 [4]. Barkhordari et al. further demonstrated that interpretable XGBoost models are relevant to JECE’s environmental-remediation scope by predicting heavy-metal removal efficiency in electrokinetic remediation [7]. These studies establish that ML is useful for environmental treatment modeling.

The present study adds a validation-layer contribution: adsorption ML workflows should explicitly state whether they are predicting within a known source, interpolating within a compiled dataset, transferring to unseen adsorbent names, or transferring to unseen sources. This distinction is critical because high row-wise R^2 can be overoptimistic when the deployment target is a new laboratory, adsorbent synthesis route, contaminant matrix, or literature source. The broader ML literature has identified leakage and validation mismatch as major reproducibility risks [9, 10], and adsorption datasets have several mechanisms by which such mismatch can occur.

4.5. Practical implications for adsorption model deployment

For practical adsorption screening, a single universal RE model should be used cautiously. A more defensible workflow is to define the pH-concentration applicability domain before modeling, exclude or separately model poorly represented regimes, use source-specific or laboratory-calibrated models for high-accuracy optimization within a known experimental system, use capacity-free screening models for preliminary adsorbent and operating-condition selection, attach an applicability-domain trust score to each screening prediction, and report

source-held-out validation when predictions are intended for new adsorbents,
610 contaminants, laboratories, or literature domains. This framing allows high- R^2
models to remain useful without overstating their generality.

The most direct route to improving cross-source performance is not simply
replacing XGBoost or LightGBM with a more complex learner. The source au-
dit suggests that external performance would more likely improve through richer
615 mechanistic descriptors, better source metadata, and validation-aware modeling.
Future versions of the dataset should prioritize adsorbent surface chemistry, syn-
thesis and activation conditions, solution chemistry, and explicit laboratory/source
labels. Once those descriptors are available, hierarchical or multi-task models
could separate universal adsorption trends from source-specific offsets, while still
620 reporting reference-held-out or laboratory-held-out validation as the primary de-
ployment test.

4.6. *Limitations and future work*

This study has several limitations. First, the dataset represents adsorption
tests for AMD-relevant metal and metalloid contaminants, but many records
625 are single- or limited-ion aqueous experiments rather than complete real AMD
matrices. Therefore, sulfate concentration, ionic strength, competing ions,
dissolved organic matter, and site-specific mine-water composition should be
added before claiming direct deployment to complex field AMD. Second, the
dataset does not contain explicit laboratory-origin labels for all records; bibli-
630 ographic reference was therefore used as a source proxy. Future work should
annotate records as in-house, external-laboratory, or literature-derived and repeat
validation with leave-laboratory-out splits. Third, the ion descriptor table should
be expanded with fully cited descriptors such as electronegativity, hydration

energy, softness/hardness, and likely aqueous speciation. Fourth, important
635 adsorbent descriptors such as pH_{pzc} , pore volume, pore size, zeta potential,
FTIR-derived functional groups, and cation-exchange capacity are incomplete in
the present dataset. Fifth, the neutral/low-concentration regime contained only 26
observations and was excluded from primary modeling; predictions in that regime
should therefore be treated as outside the present applicability domain until more
640 data are available. Sixth, because all records were literature-derived, reference-
held-out validation should be interpreted as a proxy for external-source transfer,
not as a substitute for prospective laboratory validation. Finally, the dataset was
designed as a heterogeneous ML benchmark for source-aware validation rather
than as a complete systematic meta-analysis of all AMD treatment or heavy-metal
645 adsorption studies. These limitations define clear routes for improving external
predictive performance while preserving the source-aware validation principle.

5. Conclusions

This study developed a source-aware and leakage-aware validation framework
plus applicability-domain audit for AMD-relevant metal-contaminant adsorption
650 removal-efficiency prediction. It should not be read as a claim of a state-of-the-art
universal adsorption predictor. High R^2 values were achievable within individual
adsorption sources, with source-specific XGBoost models reaching R^2 up to
0.992. Because the neutral/low-concentration regime contained only 26 observa-
tions, primary global modeling excluded that sparse regime and used a 537-record
655 restricted domain. Within this domain, random row-wise validation reached
 $R^2 = 0.806$ for the full capacity LightGBM process model, adsorbent-held-out
validation reached $R^2 = 0.532$ for the capacity-free numeric expanded-LightGBM

screening model, and reference-held-out validation reached $R^2 = 0.330$ for the full-capacity expanded-LightGBM sensitivity model. The best capacity-free
660 reference-held-out model reached $R^2 = 0.305$. A pre-experiment applicability-domain trust score then reduced capacity-free reference-held-out MAE from 18.67 to 14.53 at 40% prediction coverage and identified pH- C_0 regions where additional experiments would most directly improve future screening models. These findings show that high-accuracy interpolation, applicability-domain
665 definition, and cross-source generalization are different scientific tasks. High adsorption-ML accuracy is meaningful when the claim is interpolation within known sources or well-represented operating domains, but it is not sufficient by itself to claim deployment to unseen adsorbents, unseen studies, or new laboratories. Future adsorption ML studies should explicitly state the validation
670 question being answered, separate capacity-inclusive process-monitoring models from capacity-free screening models, attach reliability scores to prospective screening predictions, and include source- or laboratory-held-out validation when deployment beyond the training source is claimed.

CRedit authorship contribution statement

675 Yash Bisht: Conceptualization, Methodology, Software, Formal analysis, Visualization, Writing – original draft. Margreth Tadie: Supervision, Validation, Writing – review and editing. Susmit Chitransh: Data curation, Conceptualization, Investigation, Validation, Writing – review and editing.

Declaration of competing interest

680 The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This research did not receive any specific grant from funding agencies in the
685 public, commercial, or not-for-profit sectors.

Data availability

The curated dataset, analysis scripts, and generated benchmark outputs are available from the authors upon reasonable request and will be deposited in a public repository before final publication.

690 Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used OpenAI’s Codex to assist with code organization, LaTeX formatting, and language polishing. After using this tool, the authors reviewed and edited the content as needed and take full
695 responsibility for the content of the manuscript.

Acknowledgements

The authors acknowledge the institutional support of the Indian Institute of Technology Roorkee and Stellenbosch University.

References

- 700 [1] G.S. Simate, S. Ndlovu, Acid mine drainage: Challenges and opportunities, *Journal of Environmental Chemical Engineering* 2 (2014) 1785–1803. <https://doi.org/10.1016/j.jece.2014.07.021>.
- [2] V. Dube, Z. Phiri, A.T. Kuvarega, B.B. Mamba, L.-A. de Kock, Exploring acid mine drainage treatment through adsorption: a bibliometric analysis, *Environmental Science and Pollution Research* 31 (2024) 59659–59680. <https://doi.org/10.1007/s11356-024-35047-2>.
705
- [3] N. Hafsa, S. Rushd, M. Al-Yaari, M. Rahman, A generalized method for modeling the adsorption of heavy metals with machine learning algorithms, *Water* 12 (2020) 3490. <https://doi.org/10.3390/w12123490>.
- 710 [4] C. Lu, Z. Xu, B. Dong, Y. Zhang, M. Wang, Y. Zeng, C. Zhang, Machine learning for the prediction of heavy metal removal by chitosan-based flocculants, *Carbohydrate Polymers* 285 (2022) 119240. <https://doi.org/10.1016/j.carbpol.2022.119240>.
- [5] C. Wang, Y. Zhao, Y. Gao, H. Chen, X. Li, B. Zhou, D. Fan, Z. Fang, J. Liu, Interpretable machine learning for predicting heavy metal removal and optimizing biochar characteristics, *Journal of Water Process Engineering* 68 (2024) 106484. <https://doi.org/10.1016/j.jwpe.2024.106484>.
715
- [6] Z.M. Yaseen, F.L. Alhalimi, Heavy metal adsorption efficiency prediction using biochar properties: a comparative analysis for ensemble machine learning models, *Scientific Reports* 15 (2025). <https://doi.org/10.1038/s41598-025-96271-5>.
720

- 725 [7] M.S. Barkhordari, N. Zhou, K. Li, C. Qi, Interpretable machine learning for predicting heavy metal removal efficiency in electrokinetic soil remediation, *Journal of Environmental Chemical Engineering* 12 (2024) 114330. <https://doi.org/10.1016/j.jece.2024.114330>.
- [8] S.M. Lundberg, S.-I. Lee, A unified approach to interpreting model predictions, in: *Advances in Neural Information Processing Systems* 30, 2017, pp. 4765–4774.
- 730 [9] S. Kapoor, A. Narayanan, Leakage and the reproducibility crisis in machine-learning-based science, *Patterns* 4 (2023) 100804. <https://doi.org/10.1016/j.patter.2023.100804>.
- [10] J.-J. Zhu, M. Yang, Z.J. Ren, Machine learning in environmental research: Common pitfalls and best practices, *Environmental Science & Technology* 57 (2023) 17671–17689. <https://doi.org/10.1021/acs.est.3c00026>.
- 735 [11] H.V. Patel, J. Green, H. Park, S. Luster-Teasley Pass, R. Zhao, A hybrid response surface methodology and machine learning framework for quantifying effects of physicochemical parameters on PFAS distribution, *ACS ES&T Water* (2026) Advance online publication. <https://doi.org/10.1021/acsestwater.5c01162>.
- 740 [12] T. Chen, C. Guestrin, XGBoost: A scalable tree boosting system, in: *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, ACM, San Francisco, CA, USA, 2016, pp. 785–794. <https://doi.org/10.1145/2939672.2939785>.

- [13] J.H. Friedman, Greedy function approximation: A gradient boosting machine, *Annals of Statistics* 29 (2001) 1189–1232. <https://doi.org/10.1214/aos/1013203451>.
745
- [14] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, T.-Y. Liu, LightGBM: A highly efficient gradient boosting decision tree, in: *Advances in Neural Information Processing Systems* 30, 2017, pp. 3146–3154.
- [15] L. Breiman, Random forests, *Machine Learning* 45 (2001) 5–32. <https://doi.org/10.1023/A:1010933404324>.
750
- [16] P. Geurts, D. Ernst, L. Wehenkel, Extremely randomized trees, *Machine Learning* 63 (2006) 3–42. <https://doi.org/10.1007/s10994-006-6226-1>.
- [17] Z. Shen, Y. Zhang, O. McMillan, F. Jin, A. Al-Tabbaa, Characteristics and mechanisms of nickel adsorption on biochars produced from wheat straw pellets and rice husk, *Environmental Science and Pollution Research* 24 (2017) 12809–12819. <https://doi.org/10.1007/s11356-017-8847-2>.
755
- [18] L.-Y. Gao, J.-H. Deng, G.-F. Huang, K. Li, K.-Z. Cai, Y. Liu, F. Huang, Relative distribution of Cd(II) adsorption mechanisms on biochars derived from rice straw and sewage sludge, *Bioresource Technology* 272 (2019) 114–122. <https://doi.org/10.1016/j.biortech.2018.09.138>.
760
- [19] E.F. Zama, Y.-G. Zhu, B.J. Reid, G.-X. Sun, The role of biochar properties in influencing the sorption and desorption of Pb(II), Cd(II) and As(III) in aqueous solution, *Journal of Cleaner Production* 148 (2017) 127–136. <https://doi.org/10.1016/j.jclepro.2017.01.125>.
765

- [20] X. Cui, S. Fang, Y. Yao, T. Li, Q. Ni, X. Yang, Z. He, Potential mechanisms of cadmium removal from aqueous solution by *Canna indica* derived biochar, Science of the Total Environment 562 (2016) 517–525. <https://doi.org/10.1016/j.scitotenv.2016.03.248>.

770